# Intelligent Cyber Crime Reporting System

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# Problem Statement Number:SIH25132

Organization: AICTE.
Category :Software

## Problem Description:

- Cyber crimes such as online fraud, phishing, identity theft, and social engineering attacks are increasing rapidly in both volume and complexity. Although online cyber crime reporting portals exist, they primarily focus on complaint registration and storage. These systems lack intelligent mechanisms to analyze newly registered complaints, correlate them with previous similar cases, and assist law enforcement officials with insights such as crime patterns, modus operandi, and resolution history.
- As a result, investigators often spend significant time manually searching past records, leading to delayed investigations, repeated patterns of crime, and slower case resolution. Additionally, citizens lack access to real-time awareness dashboards that inform them about emerging cyber crime trends and newly evolving attack techniques.

# Objectives

- To design a centralized cyber crime reporting platform for citizens and officials.
- To implement AI and Machine Learning techniques for analyzing complaint descriptions.
- To identify and display similar past cyber crime cases using similarity scoring.
- To assist law enforcement by presenting modus operandi, crime patterns, and case resolution summaries from historical data.
- To provide citizens with an awareness dashboard showing new and emerging cyber crime trends.
- To reduce investigation time and improve decision-making efficiency for officials.
- To ensure data privacy, role-based access, and ethical use of AI.

## Background

Several cyber crime reporting systems exist at national and international levels that allow users to file complaints and track their status; however, most of these platforms function as static databases and do not leverage advanced analytics or artificial intelligence for deeper insight generation. Recent research has explored areas such as NLP-based crime text classification, machine learning-driven fraud detection, crime trend visualization, and threat intelligence platforms in cybersecurity, demonstrating the potential of AI in crime analysis.

# Background and Related work(Contt..)

## Related Work

Despite this, these solutions are typically limited to isolated classification or prediction tasks and fail to integrate complaint similarity matching, modus operandi analysis, and investigative assistance within a single framework. The selected project title reflects the integration of AI-driven analysis, cyber crime reporting, intelligent case correlation, and decision-support for law enforcement, thereby addressing the identified gaps and justifying the need for a unified intelligent solution.

# Analysis of Problem Statement

- **Existing System Limitations:**
- Manual analysis of complaints by officials
- No automatic detection of similar cases
- Lack of crime pattern and modus operandi insights
- Minimal public awareness beyond advisories
- Delayed investigation and response times

- **Identified Gaps:**
- Absence of AI-based similarity matching
- No real-time analytical dashboards
- No historical learning from solved cases
- Limited decision support for investigators

# Analysis of Problem Statement(Contt..)

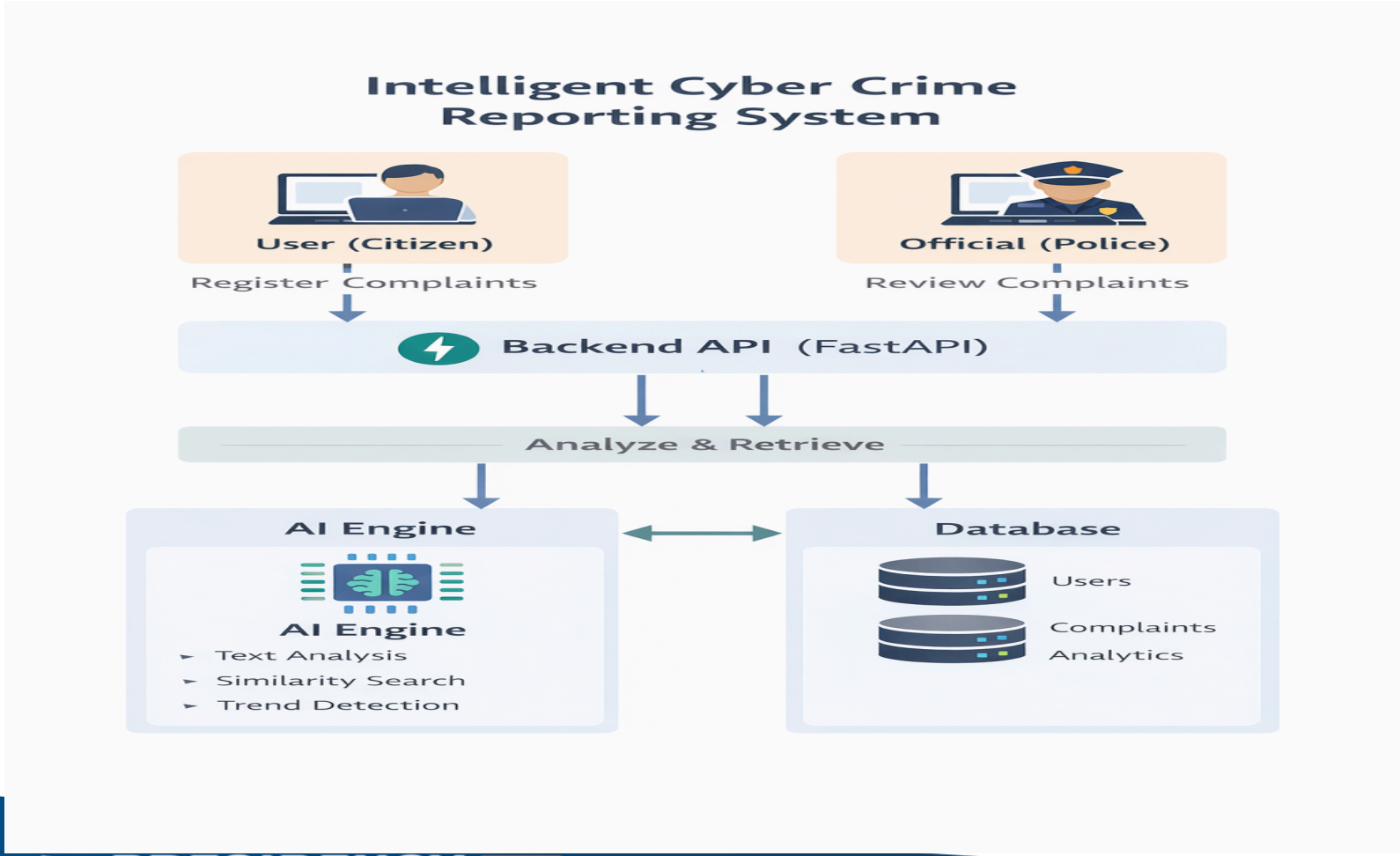
- **Proposed System Capabilities:**
- Automated text analysis using NLP
- Similarity detection using ML models
- Pattern extraction from historical data
- Dual dashboards (Officials & Citizens)
- Faster, informed, and data-driven investigations

# Innovation or Novel Contributions

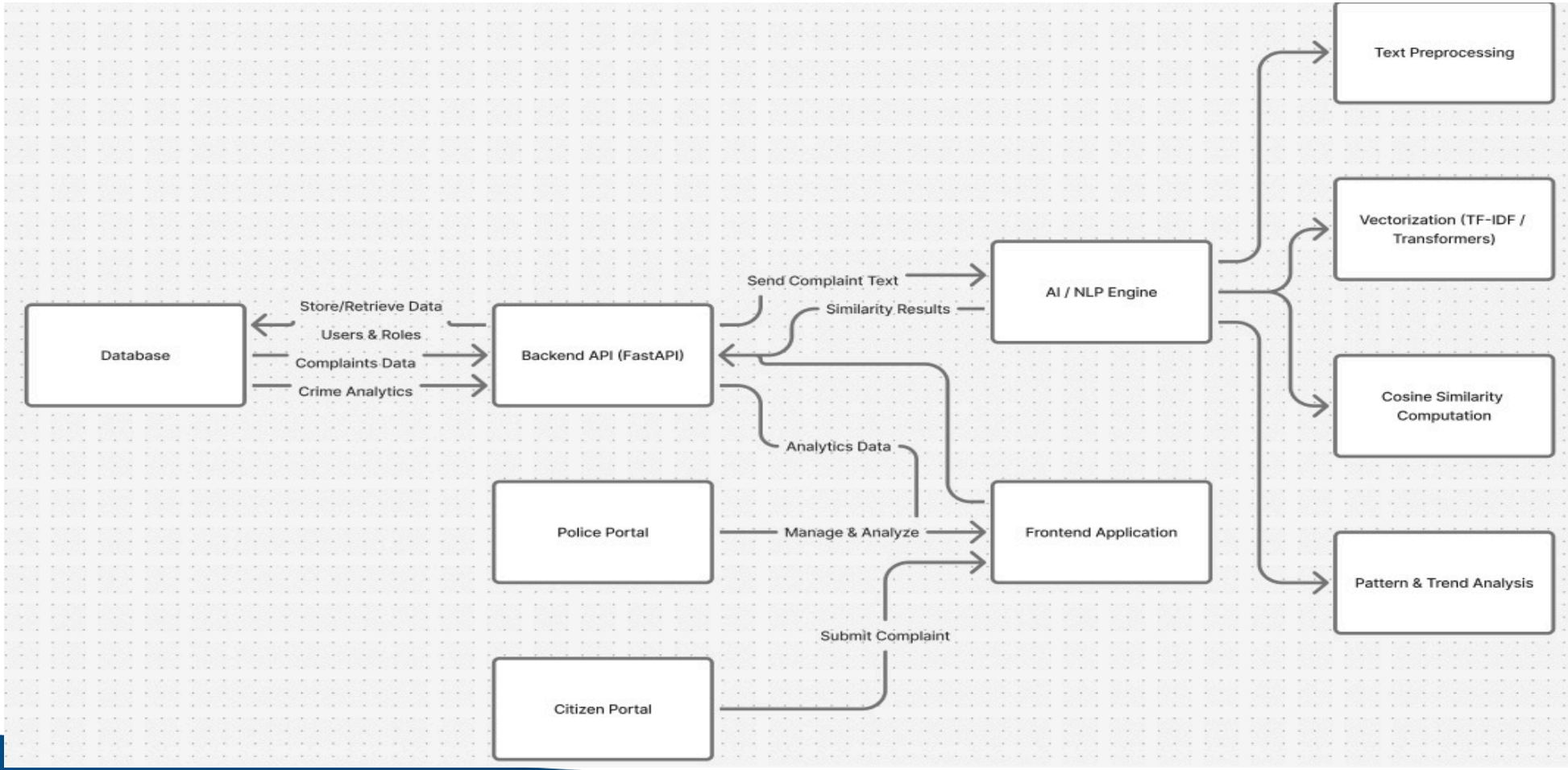
**The proposed system introduces the following innovative contributions:**

- AI-Based Complaint Similarity Matching
- Automatically matches new complaints with past incidents using similarity scores.
- Modus Operandi Identification
- Highlights common attack techniques used in similar crimes to assist investigators.
- Decision-Support System for Law Enforcement
- Provides insights from historical case resolutions instead of just raw data.
- Citizen Awareness Dashboard
- Educates users about new and emerging cyber crime patterns in real time.
- Learning-Based System Architecture
- System improves accuracy as more complaint data is added over time.
- Ethical AI Usage
- Acts as a recommendation and analysis tool, not an automated decision-maker.

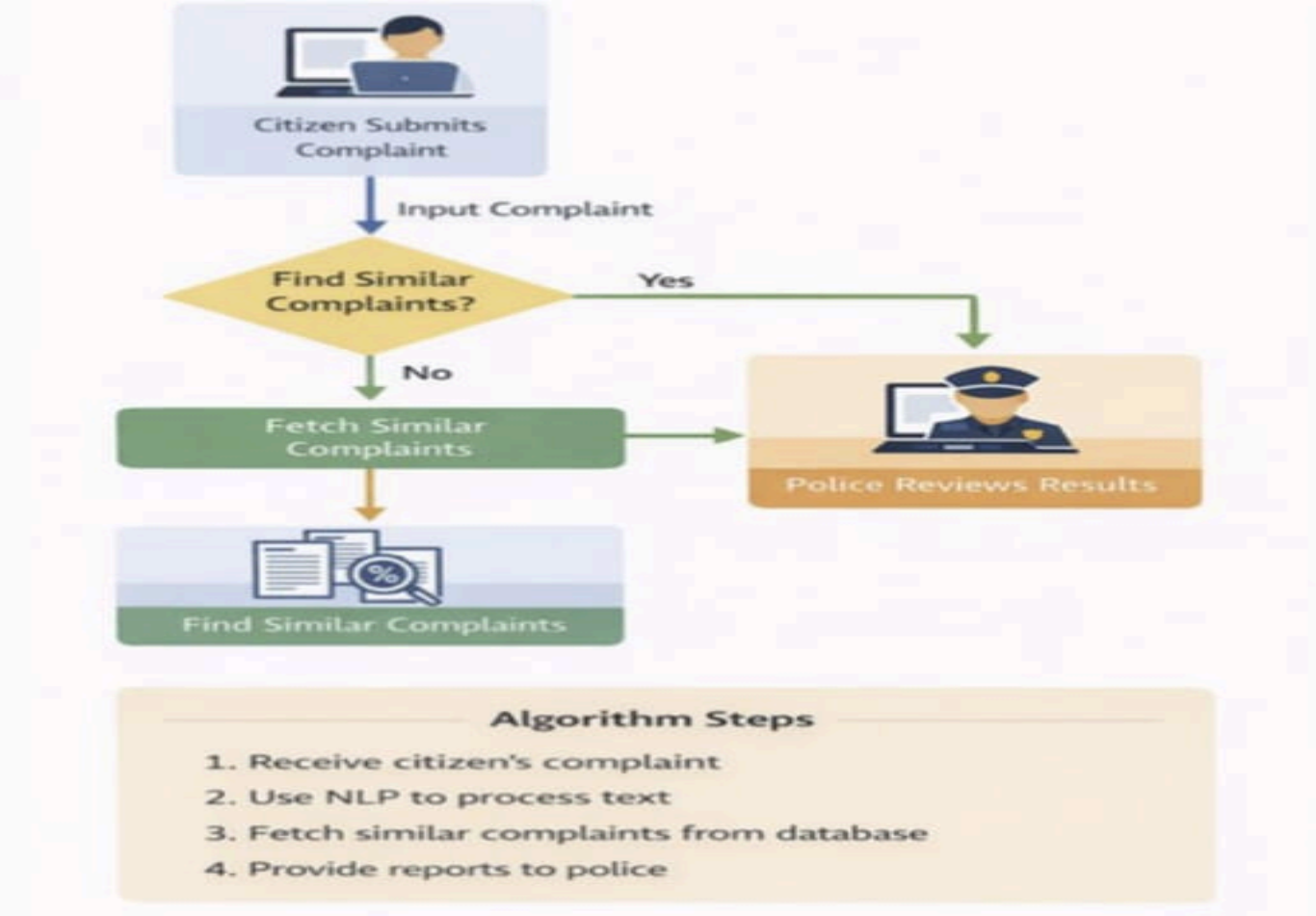
# Architecture Diagram



# BLOCK DIAGRAM



# ALOGORITHM


Citizen Submits Complaint
Input Complaint
Find Similar Complaints?
Yes
No
Fetch Similar Complaints
Police Reviews Results
Find Similar Complaints
Algorithm Steps
1. Receive citizen's complaint
2. Use NLP to process text
3. Fetch similar complaints from database
4. Provide reports to police

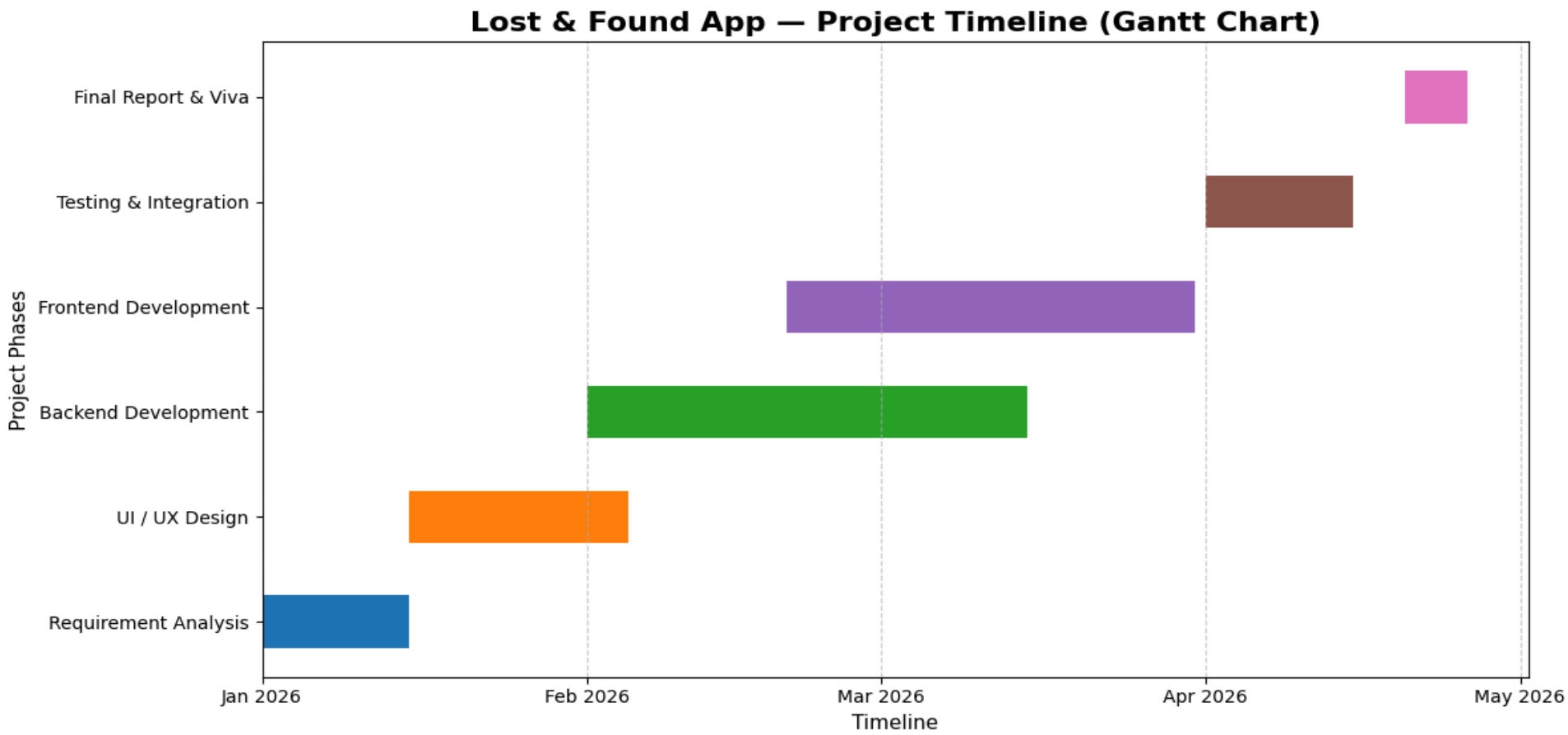
# Github Link

The Github link provided should have public access permission.

## Github Link

https://github.com/Yagneshvss/cybercrimereporting-system/tree/main

# Timeline of the Project (Gantt Chart)



# References (IEEE Paper format)


**[1]Tan, S. Y., & Chong, C. R.** (2023). *An Effective Lost and Found System in University Campus*. Journal of Information System and Technology Management, 8(32), 99-112.
— Describes the design and implementation of a web/mobile lost and found system for a university campus.

**[2]Kumar, S., Bhatnagar, D., Gahlot, D., & Gola, B. S.** (2022). *Web-Based Lost and Found System*. MIT International Journal of Computer Science and Information Technology, Vol. 11, No. 1.
— Presents a web application for lost and found item management with user registration and item listing features.

**[3]CampusTrace: Application for Lost and Found Items on Campus** (2024). TCSST Journal.
— A study on a campus-oriented lost and found platform that improves reporting and item return processes.

**[4]Dutta, A., Tolani, Y., & Singhal, N.** (2024). *Automated Lost and Found System with Peer-to-Peer Communication*. JETIR, 11(5).
— Discusses an automated lost and found system using peer-to-peer communication to connect finders and owners efficiently.

**[5]Zhou, M., Fung, I., Yang, L., et al.** (2023). *LostNet: A Smart Way for Lost and Find*. arXiv preprint.
— Proposes an image-based retrieval mechanism to match lost and found items using machine-learning-driven features.

Thank
You!